

RAHUL VUDATHU

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[Portfolio](#) ♦ [LinkedIn](#) ♦ [LeetCode](#) ♦ [GitHub](#)

OBJECTIVE

Proactive final-year B.Tech Computer Science student at VIT-AP University, eager to apply technical skills in real-world projects. Passionate about learning, problem-solving, and contributing to collaborative, fast-paced environments.

EDUCATION

B.Tech in Computer Science and Engineering, Vellore Institute of Technology - Amaravathi 2022 - 2026
CGPA: 8.89

Intermediate, Sri Chaitanya Junior College 2020 - 2022
Percentage: 94.1

SKILLS

Technical Skills: Python, Java, Spring, Spring Boot (Backend), Web Development, ReactJS, Bootstrap, Tailwind CSS, TensorFlow, Machine Learning, Deep Learning, Data Structures

Databases: SQL, Database Management Systems (DBMS)

Data Science Analytics: Pandas, NumPy, scikit-learn, Matplotlib, Data Visualization

Development Tools: Git, GitHub, VSCode, JIRA, Jupyter Notebook

Soft Skills: Communication, Team Collaboration, Adaptability, Critical Thinking, Problem-Solving

PROJECTS

Invoice Generator Web App *Spring Boot, Spring Security (Clerk), ReactJS, Bootstrap, MongoDB, Spring JPA, SMTP, HTML2PDF*

A full-stack invoice generation platform with secure Clerk-based authentication. Allows users to choose from multiple professional invoice templates, download them as PDFs, and email them using backend SMTP. MongoDB (accessed via Spring Data MongoDB JPA) is used for storing user-specific invoice data and dashboard statistics. Includes a personalized dashboard for tracking total billed and pending amounts. [GitHub](#)

Limit – Task Manager

React, Tailwind CSS, Local Storage, Framer Motion

A comprehensive to-do app with Upcoming, Ongoing, and Completed task sections. Features include task categorization, smooth transitions, reminders, and team collaboration. [GitHub](#)

CO2 Emission Prediction *Python, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, RandomForest, SVM, Linear Regression* Developed a machine learning pipeline to predict CO2 emissions of countries from 1960 to 2015 using key features like GDP, population, and industrial indicators. Utilized models such as Random Forest, SVR, Ridge, Decision Trees, and XGBoost. Performed hyperparameter tuning with RandomizedSearchCV and evaluated performance using MAE, MSE, and RMSE. Provided detailed data visualizations using Matplotlib and Seaborn. [GitHub](#)

Expense Tracker App

React, Context API, Chart.js, Bootstrap

A modern and responsive web app for managing personal finances. Features include modules like **Transactions**, **Analytics**, and **Settings**, along with a personalized dashboard to visualize income and expense trends. Users can export their data as a structured `.json` file and optionally re-upload it to restore previous sessions. Real-time updates and dynamic charts provide insightful spending analysis. [GitHub](#) · [Live](#)

Frozen Lake RL Solver

Python, OpenAI Gym, NumPy, Matplotlib

Designed and developed a reinforcement learning agent to solve OpenAI Gym's **FrozenLake-v1** environment using both **Q-Learning** and **Policy-Based Learning**. The agent learns optimal navigation strategies in a stochastic 4x4 grid world with slippery dynamics. Key features include **ϵ -greedy exploration** and **gradient-based policy optimization**. Comparative experiments analyze convergence speed, stability, and reward accumulation across

both approaches. Real-time performance tracking and episode-wise plotting were used to evaluate learning progress.
[GitHub](#)

CERTIFICATIONS

- **GitHub Foundations** - [GitHub](#)
- **Oracle Cloud Infrastructure 2024 Generative AI Certified Professional** - [Oracle](#)
- **MongoDB Associate Database Administrator** - [MongoDB](#)
- **AWS Academy Graduate - AWS Academy Cloud Architecting** - [AWS](#)
- **GitHub Administration** - [GitHub](#)
- **React (Basic)** – [HackerRank](#)
- **Geekathon 1.0 – Hackathon by GFG VIT-AP** – [GeeksforGeeks](#)

EXTRA-CURRICULAR ACTIVITIES

Innovators Quest – VIT-AP	
2023–2024	Web Wizards Co-Lead
• Conducted workshops on modern web development (ReactJS, Bootstrap). • Mentored junior developers and contributed to internal project UIs. • Played a key role in organizing technical sessions and outreach.	
2024–2025	Technical Team Lead
• Leading cross-functional tech teams on club projects and events. • Oversee end-to-end technical architecture and execution. • Collaborate with other domains to drive innovation and mentorship.	